7.1 Normal vector and Cartesian equation

Definition and results

A non-zero normal vector to a line is orthogonal to every direction vector of that line. If $\vec{n} = (a; b) \neq (0; 0)$ is normal and $A(x_A; y_A)$ belongs to d, then d has equation $a(x - x_A) + b(y - y_A) = 0$, i.e. $ax + by + c = 0$. A possible direction vector is $(-b; a)$.

Status and justification. $M \in d$ is equivalent to $\overrightarrow{AM} \cdot \vec{n} = 0$. In an orthonormal coordinate system, expanding this product gives the equation. Conversely, the set thus defined is the line through A orthogonal to n.

Worked example. Via $A(1; 2)$ and normal $(2; -3)$: $2(x - 1) - 3(y - 2) = 0$, so $2x - 3y + 4 = 0$. The vector (3; 2) directs the line: product $6 - 6 = 0$.

Watch out

$(a; b)$ is normal, and in general not a direction vector. $(a; b)$ must not be zero.

Immediate application. Equation of the line through $(2; -1)$ with normal vector (3; 4).

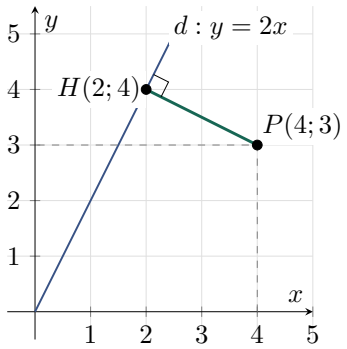
7.2 Orthogonal projection and distance

Definition and results

The projection H of P onto d is characterised by $H \in d$ and $\overrightarrow{PH}$ collinear with a normal vector of d. For $d : ax + by + c = 0$, write $H = P - \lambda(a; b)$, with $\lambda = \dfrac{ax_P + by_P + c}{a^2 + b^2}$. The distance is $PH = \dfrac{|ax_P + by_P + c|}{\sqrt{a^2 + b^2}}$. The distance formula is a derived extra, not a stand-alone routine.

Status and justification. Substitute $x_H = x_P - \lambda a$, $y_H = y_P - \lambda b$ into the equation: $ax_P + by_P + c - \lambda(a^2 + b^2) = 0$. The denominator is positive; we obtain H, then $PH = |\lambda| \sqrt{a^2 + b^2}$.

Worked example. For $P(3; 4)$ and $d : x + y - 2 = 0$, $\lambda = \dfrac{3 + 4 - 2}{2} = \dfrac{5}{2}$, so $H(\dfrac{1}{2}; \dfrac{3}{2})$ and $PH = \dfrac{5}{\sqrt{2}} = \dfrac{5\sqrt{2}}{2}$. Check $x_H + y_H = 2$.



The segment PH is perpendicular to d. The two axes have the same scale; the coordinates of P and H can be read directly.

Watch out

A distance is positive or zero: keep the absolute value. Two checks, membership and perpendicularity, confirm the projection.

Immediate application. Project $P(4; 1)$ onto $d : x = 2$.

7.3 Equation of a circle

Definition and results

The circle with centre $I(\alpha; \beta)$ and radius $r > 0$ has equation $(x - \alpha)^2 + (y - \beta)^2 = r^2$. A equation $x^2 + y^2 + ux + vy + w = 0$ is handled by completing both squares. If the squared radius obtained is negative, the set is empty; if it is zero, it is a point.

Status and justification. M belongs to the circle if and only if $IM = r$. Both sides being positive or zero, we can square and use the distance formula in an orthonormal coordinate system.

Worked example. $x^2 + y^2 - 4x + 6y + 9 = 0$ becomes $(x - 2)^2 + (y + 3)^2 = 4$. Centre $(2; -3)$, radius 2. The signs of the centre's coordinates are opposite to those in the brackets.

Watch out

A quadratic equation in x and y is not always a circle: check the coefficients and that the real radius.

Immediate application. Identify $x^2 + y^2 + 2x - 8y + 17 = 0$.

7.4 Solving a configuration

Definition and results

An intersection is found by solving the equations simultaneously. For a circle and a line parallel to an axis, substituting the constant coordinate gives a quadratic equation. For a perpendicular bisector, write the equality of the squared distances; the quadratic terms cancel out.

Status and justification. The equivalences rely on membership of both sets. The equality $MA = MB$ is equivalent to $MA^2 = MB^2$, since the distances are positive.

Worked example. The circle $x^2 + y^2$ = 25 and the line x = 3 meet at (3; 4) and (3; -4). The perp. bis. of A(0; 0) and B(4; 2) satisfies $x^2 + y^2 = (x-4)^2 + (y-2)^2$, so 2 $x + y = 5$.

Watch out

An intersection is not the union of the solution sets. Check the points in both equations.

Immediate application. Intersect ($x-1)^2 + (y+2)^2$ = 9 with $y = 1$.